

NETWORKS AND COMPLEXITY

Solution 16-3

This is an example solution from the forthcoming book Networks and Complexity.

Find more exercises at <https://github.com/NC-Book/NCB>

Ex 16.3: Abstract sensitivity example [Lvl. 2]

You study an ODE that is quite complicated,

$$\dot{x} = -x^4 + 2px^3 + 2p^2x - px^2 + x - 2p$$

However, you know that in the real world system $p = 1$, $x = 2$.

- a) Verify that $x^* = 2$ is indeed a steady state at $p = 1$.

Solution

Substituting $p = 1$ yields

$$\dot{x} = -x^4 + 2x^3 + 2x - x^2 + x - 2 \quad (1)$$

and then substituting $x = 2$ we find

$$\dot{x} = -2^4 + 2 \cdot 2^3 + 2 \cdot 2 - 2^2 + 2 - 2 = 0$$

so this is indeed a steady state.

- b) Use the implicit function theorem to estimate how the steady state will respond if p is changed a little bit.

Solution

We need

$$f_x(x^*, p^*) = \left. \frac{\partial}{\partial x}(-x^4 + 2px^3 + 2p^2x - px^2 + x - 2p) \right|_* \quad (2)$$

$$= (-4x^3 + 6px^2 + 2p^2 - 2px + 1)|_* \quad (3)$$

$$= -4 \cdot 2^3 + 6 \cdot 1 \cdot 2^2 + 2 \cdot 1^2 - 2 \cdot 1 \cdot 2 + 1 \quad (4)$$

$$= -32 + 24 + 2 - 4 + 1 \quad (5)$$

$$= -9 \quad (6)$$

and also

$$f_p(x^*, p^*) = \left. \frac{\partial}{\partial p}(-x^4 + 2px^3 + 2p^2x - px^2 + x - 2p) \right|_* \quad (7)$$

$$= (2x^3 + 4px - x^2 - 2)|_* \quad (8)$$

$$= 2 \cdot 2^3 + 4 \cdot 1 \cdot 2 - 2^2 - 2 \quad (9)$$

$$= 16 + 8 - 4 - 2 \quad (10)$$

$$= 18 \quad (11)$$

We can now compute

$$\left. \frac{dx^*}{dp} \right|_* = -\frac{f_p}{f_x} = -\frac{18}{-9} = 2 \quad (12)$$

In words: If we start from the steady state at $p^* = 1$, $x^* = 2$ and then change p by a small amount ρ we expect the steady state to shift by 2ρ , so locally $x^* = 2 + 2\rho$.

[Bonus: Notice anything funny about the calculations in this exercise. There is a little bit more to discover here.]